

IDEAFORGE AI

AI-Powered Project Development Advisor

Project Documentation

1. TITLE PAGE

IDEAFORGE AI

AI-Powered Project Development Advisor

A Generative AI and Retrieval-Augmented Generation Based Project Planning System

Submitted by:

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Course:

B.Sc Computer Science with Artificial Intelligence

Technologies:

Python | Streamlit | Groq | RAG | Sentence Transformers | ChromaDB | ReportLab

Live Demo:

<https://ideaforge-ai-a7aiugt57ktpn352ajhb2v.streamlit.app/>

2. ABSTRACT

IdeaForge AI is an AI-powered project development advisor designed to help students and developers transform an initial project idea into a structured and feasible project plan.

The system combines a Large Language Model, Retrieval-Augmented Generation, semantic embeddings, and a vector database to provide project-specific recommendations. Users can enter a project idea and receive information such as project analysis, target users, core features, technology recommendations, difficulty level, and possible improvements.

A key feature of IdeaForge AI is its interactive project development capability. Users can continue communicating with the system through natural-language chat to modify project information, add new features, remove existing features, or refine the project.

The system uses Streamlit as the user interface, Groq with the `openai/gpt-oss-120b` model for language generation, Sentence Transformers with `all-MiniLM-L6-v2` for embeddings, ChromaDB for vector-based retrieval, and ReportLab for PDF generation.

The finalized project can be converted into a structured project blueprint and downloaded as a PDF. The system is deployed using Streamlit Community Cloud, making the application accessible through a web browser.

3. INTRODUCTION

Choosing and planning a project can be challenging, especially for students who have an idea but are unsure how to convert it into a practical implementation.

A project idea may raise several questions:

Is the idea feasible?

Who are the target users?

What features should be included?

Which technologies should be selected?

What improvements can be made?

What level of difficulty does the project have?

How should the project be structured?

Traditional approaches require students to search through multiple sources, discuss ideas with mentors, and manually prepare a project plan.

IdeaForge AI addresses this challenge by providing an AI-assisted environment for project development.

The system allows users to start with a simple idea and progressively develop it into a structured project blueprint.

4. PROJECT OVERVIEW

IdeaForge AI is a web-based AI application that works as a project development advisor.

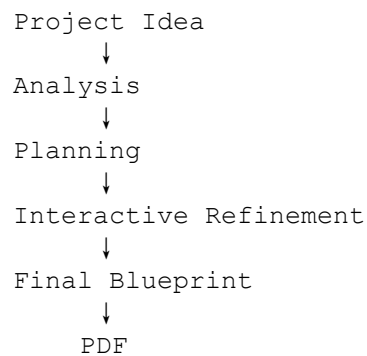
The system accepts project information from the user and uses an LLM to analyze and develop the idea.

The application provides:

- Project analysis
- Problem understanding
- Target-user identification
- Feature planning
- Technology recommendations
- Difficulty assessment
- Improvement suggestions
- Interactive project modification
- Project blueprint generation
- PDF export

The project also uses Retrieval-Augmented Generation to retrieve relevant knowledge from a vector database before generating responses.

The system is designed around the complete project-development workflow:



5. OBJECTIVES

The main objectives of IdeaForge AI are:

1. Project Idea Analysis

To analyze a user's project idea and provide structured recommendations.

2. Project Planning

To help users identify features, target users, technologies, and project requirements.

3. AI-Assisted Development

To use a Large Language Model to understand natural-language project requests.

4. Knowledge Retrieval

To use Retrieval-Augmented Generation for retrieving relevant project-related knowledge.

5. Interactive Modification

To allow users to modify project details using natural-language instructions.

6. Blueprint Generation

To create a structured representation of the final project.

7. PDF Export

To provide the final project blueprint as a downloadable PDF document.

6. PROBLEM STATEMENT

Many students have project ideas but struggle to convert those ideas into structured and feasible projects.

The main problems include:

- Difficulty in identifying appropriate project features

- Uncertainty about technology selection

Difficulty understanding project feasibility

Lack of structured project planning

Difficulty identifying target users

Time spent searching for project guidance

Repeated manual changes when project requirements evolve

Therefore, there is a need for an interactive AI-based system that can assist users throughout the project-planning process.

Problem Statement

To develop an AI-powered project development advisor that can transform a basic project idea into a structured, interactive, and downloadable project blueprint using Large Language Models, Retrieval-Augmented Generation, embeddings, and vector-based knowledge retrieval.

7. TOOLS & TECHNOLOGIES USED

Programming Language

Python

Python is used as the primary programming language for implementing the application logic, AI interaction, data processing, project management, and PDF generation.

User Interface

Streamlit

Streamlit is used to create the web-based user interface and provide interactive components such as text inputs, buttons, chat interfaces, project sections, and download functionality.

Large Language Model

Groq API

Groq provides access to the language model used by IdeaForge AI.

Model

`openai/gpt-oss-120b`

The model is responsible for understanding natural-language requests and generating project-related responses.

Embeddings

Sentence Transformers

Sentence Transformers is used to convert text into numerical vector representations.

Model

`all-MiniLM-L6-v2`

The embedding model is used for semantic similarity and retrieval.

Vector Database

ChromaDB

ChromaDB stores and retrieves vector representations of project-related knowledge.

Retrieval-Augmented Generation

RAG combines retrieved information with the user's request before sending the context to the language model.

PDF Generation

ReportLab

ReportLab is used to generate the final project blueprint as a PDF document.

Environment Variables

`python-dotenv`

Used during local development to load the Groq API key from environment variables.

Version Control

Git and GitHub

Git and GitHub are used for source-code management and version control.

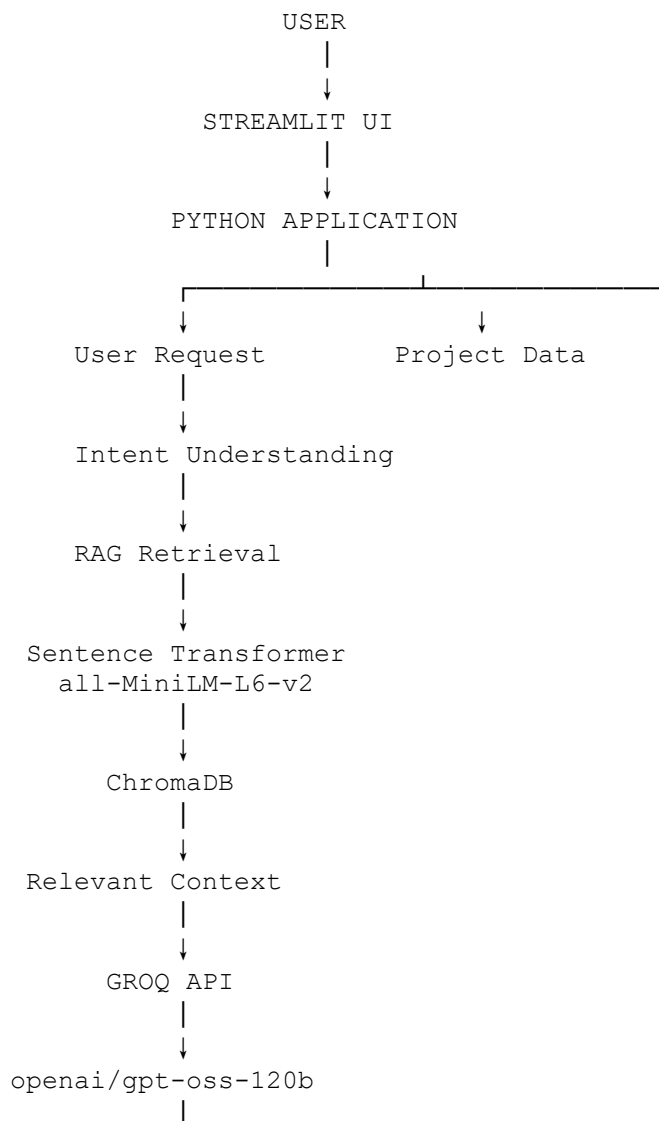
Deployment

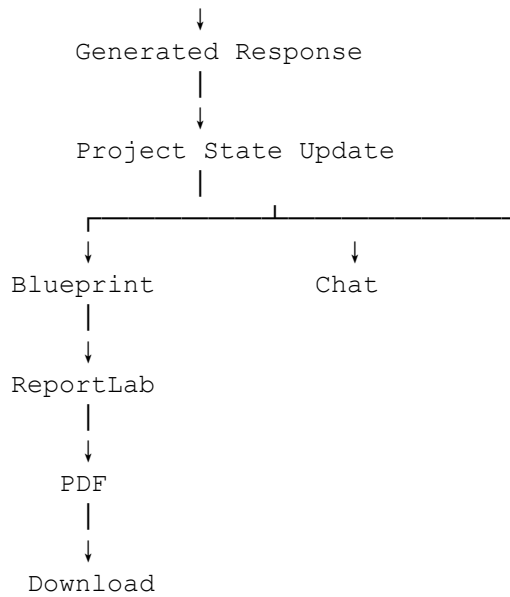
Streamlit Community Cloud

The application is deployed online using Streamlit Community Cloud.

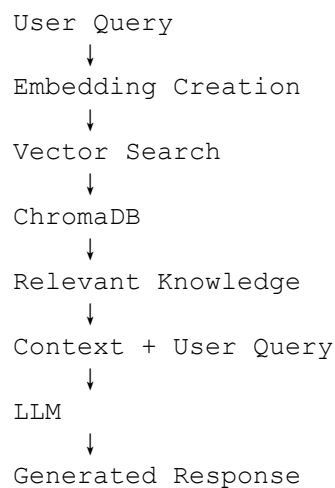
8. SYSTEM WORKFLOW / ARCHITECTURE

Overall Architecture





RAG Workflow



LLM Workflow

The Large Language Model is responsible for language understanding and generation.

The Python application remains responsible for controlling the application logic and updating project information.

9. IMPLEMENTATION

The application is implemented using Python and Streamlit.

1 Application Initialization

The application imports the required libraries and initializes the Streamlit interface.

The environment variables are loaded and the Groq client is initialized using the API key.

2 Embedding Generation

The application uses Sentence Transformers.

The embedding model is:

```
all-MiniLM-L6-v2
```

A text query is converted into a vector representation.

Conceptually:

```
Text
↓
Sentence Transformer
↓
Numerical Vector
```

3 Vector Retrieval

The generated embedding is used to query ChromaDB.

The application retrieves the most relevant stored knowledge based on semantic similarity.

4 LLM Processing

The retrieved context and user request are provided to the Groq LLM.

The model used is:

```
openai/gpt-oss-120b
```

The LLM generates the requested response.

5 Intent Understanding

When users interact through chat, the application determines what action the user wants.

Examples include:

```
Add Feature
Remove Feature
Modify Feature
Change Target Users
Change Project Information
```

6 Feature Management

If the user requests a new feature, the application processes the request.

Example:

```
User:
"Add playlist sharing."
```

↓

```
Intent:
add_feature
```

↓

```
Target:
playlist sharing
```

↓

```
Feature description generated
```

↓

```
Project updated
```

If the user requests a feature without specifying its name, the application can generate a suitable feature based on the project context.

7 Project State

The current project information is maintained using Streamlit session state.

Project information can include:

Project title

Description

Target users

Features

Technologies

Other project details

When a user modifies the project, the corresponding project state is updated.

8 Blueprint Generation

After the project is developed and finalized, the application generates a structured blueprint containing the project information.

9 PDF Generation

ReportLab is used to convert the blueprint information into a PDF document.

The generated PDF is then made available through Streamlit's download functionality.

10. SCREENSHOTS WITH EXPLANATIONS

This section should contain screenshots from the actual deployed IdeaForge AI application.

Screenshot 1 — Home / Project Input

Explanation:

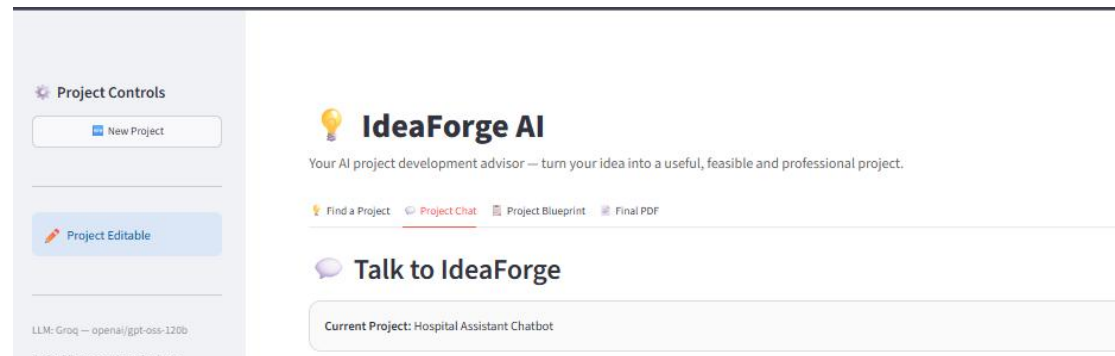
This screen allows the user to enter their project idea and provide the required project information. It acts as the starting point of the IdeaForge workflow.

The screenshot displays the 'IdeaForge AI' web application. On the left is a sidebar with 'Project Controls' containing a 'New Project' button and a 'Project Editable' button. Below this, it lists technical details: 'LLM: Gpt4o -- openai/gpt-4o-128k', 'Embeddings: nomic-embed-text', and 'RAG: ChromaDB'. The main content area features the 'IdeaForge AI' logo and tagline 'Your AI project development advisor -- turn your idea into a useful, feasible and professional project.' Below this are tabs for 'Find a Project', 'Project Chat', 'Project Blueprint', and 'Final PDF'. The 'Find a Project' tab is active, showing a form titled 'Find a Project That Fits You'. The form includes input fields for 'Your skills' (containing 'Python, Machine Learning, NLP, Streamlit, SQL'), 'Your interests' (containing 'AI, Education, Learning'), 'Preferred difficulty' (a dropdown menu set to 'Intermediate'), and 'Your project idea' (containing 'I want to create an hospital chatbot'). A red 'Analyze My Idea' button is at the bottom of the form. The top right of the interface has a 'Share' button and icons for star, refresh, and settings.

Screenshot 2 — Generated Project Analysis

Explanation:

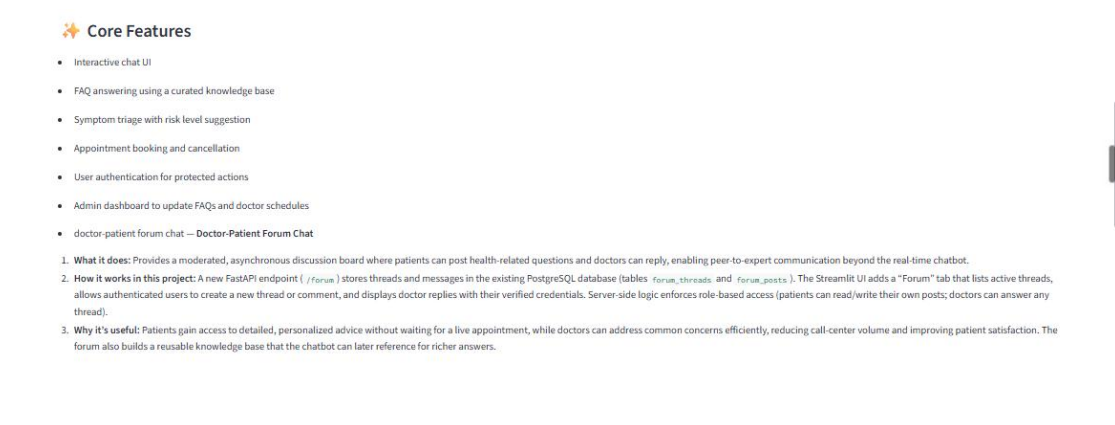
This section displays the AI-generated analysis of the project, including project understanding, target users, technologies, difficulty and other recommendations.



Screenshot 3 — Core Features

Explanation:

This section displays the core features identified for the project. Features can be modified through the interactive chat.



Screenshot 4 — Interactive Chat

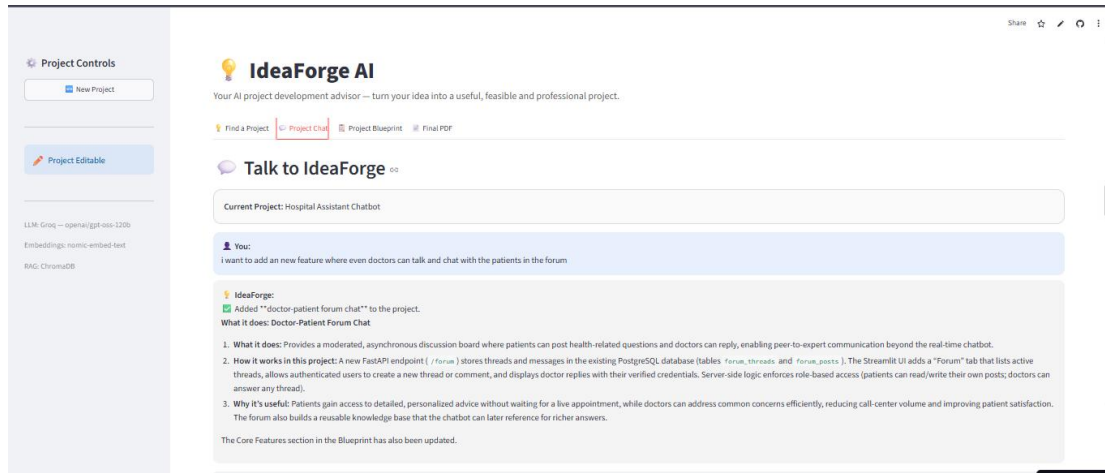
Explanation:

The chat interface allows the user to communicate naturally with IdeaForge AI and modify the project.

For example, a user can request:

“Add playlist sharing.”

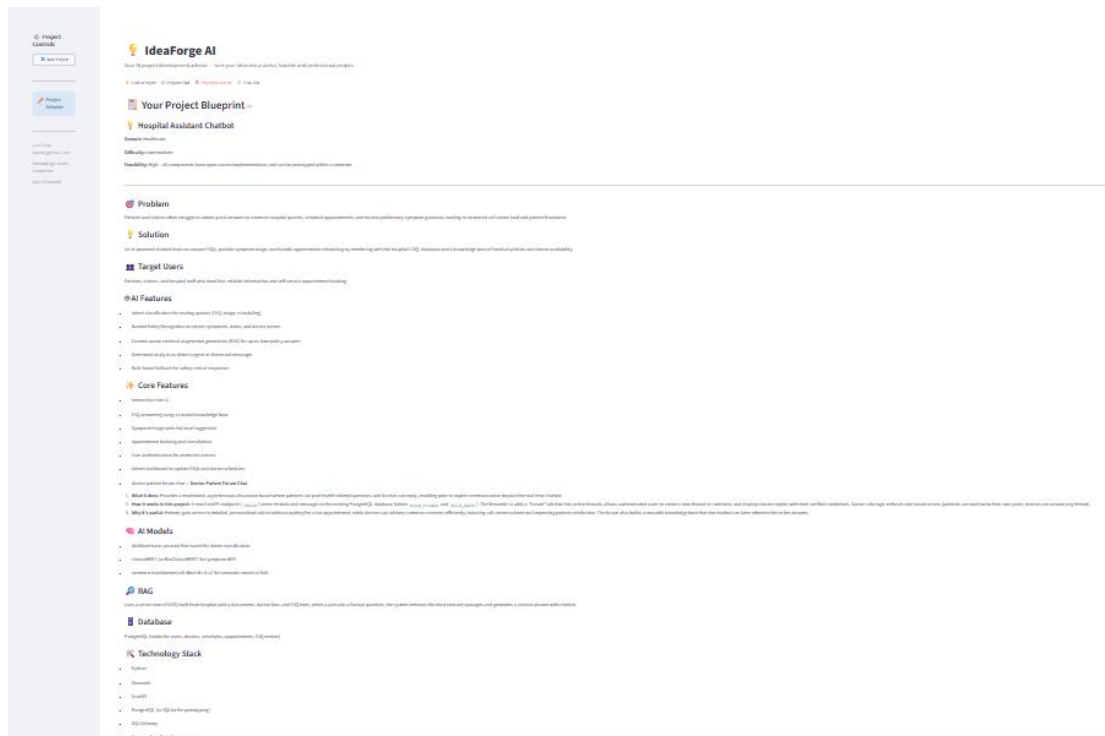
The application processes the request and updates the project.



Screenshot 5 — Project Blueprint

Explanation:

The blueprint presents the finalized project information in a structured format.



11. TESTING & TEST CASES

Testing was performed to verify the major functions of IdeaForge AI.

Test Case	Input / Action	Expected Result	Status
TC01	Enter valid project idea	Project analysis generated	Pass
TC02	Generate project blueprint	Blueprint displayed	Pass
TC03	Add a specified feature	Feature added to project	Pass
TC04	Request a new feature without specifying name	Suitable feature generated	Pass
TC05	Modify target users	Target users updated	Pass
TC06	Modify project information	Project information updated	Pass
TC07	Chat with project	AI response generated	Pass
TC08	Finalize project	Project becomes finalized	Pass
TC09	Generate PDF	PDF generated successfully	Pass
TC10	Download PDF	PDF downloaded successfully	Pass
TC11	Deploy application	Application accessible online	Pass
TC12	Groq API authentication	LLM response generated	Pass

Testing confirmed that the major project-development and deployment features operate successfully.

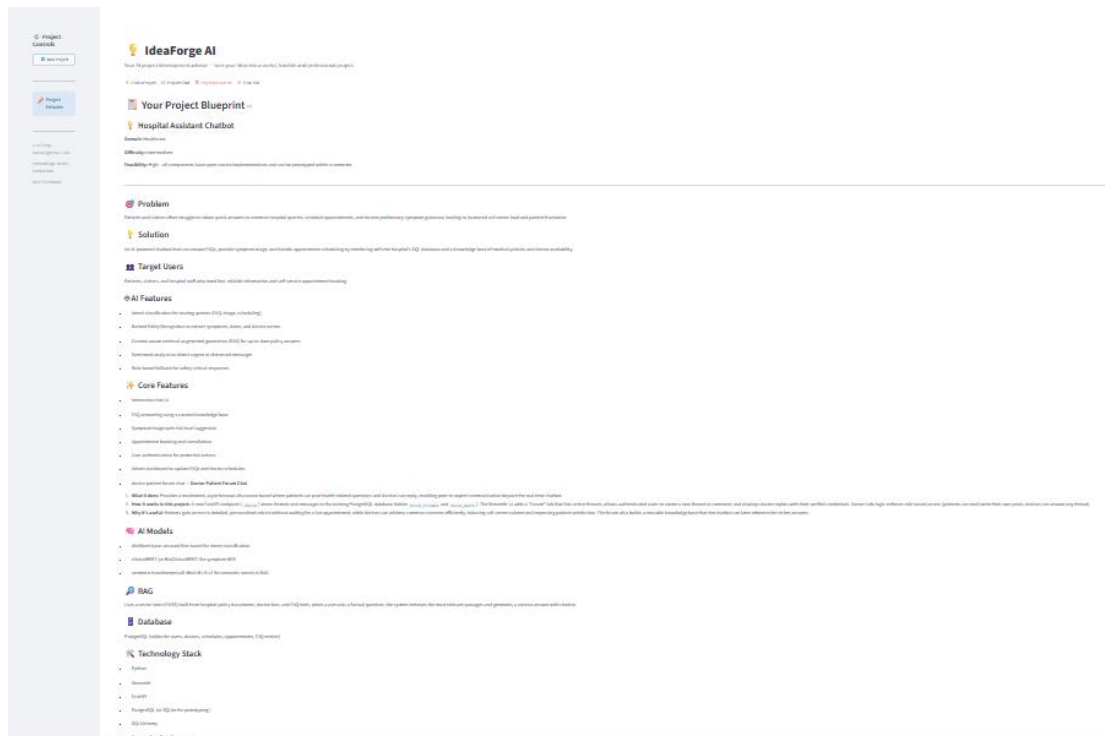
12. OUTPUT SCREENSHOTS

This section should contain screenshots showing the final outputs.

Output 1 — Generated Project

Explanation:

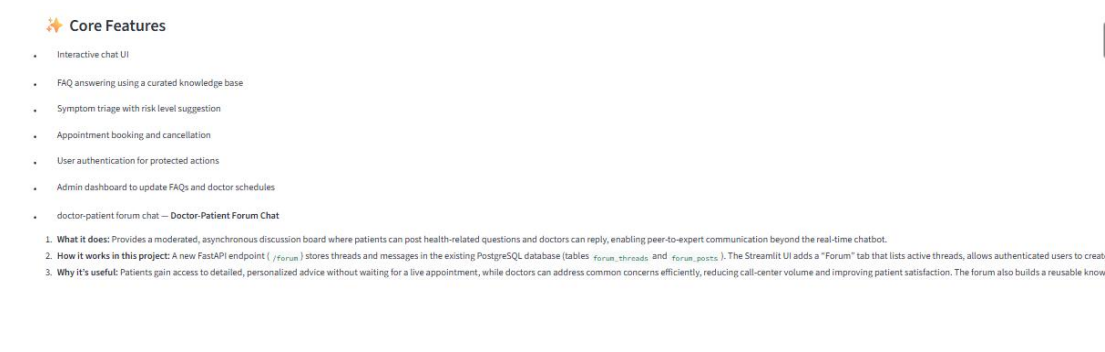
The system produces a structured project based on the user's initial idea.



Output 2 — Updated Project After Feature Addition

Explanation:

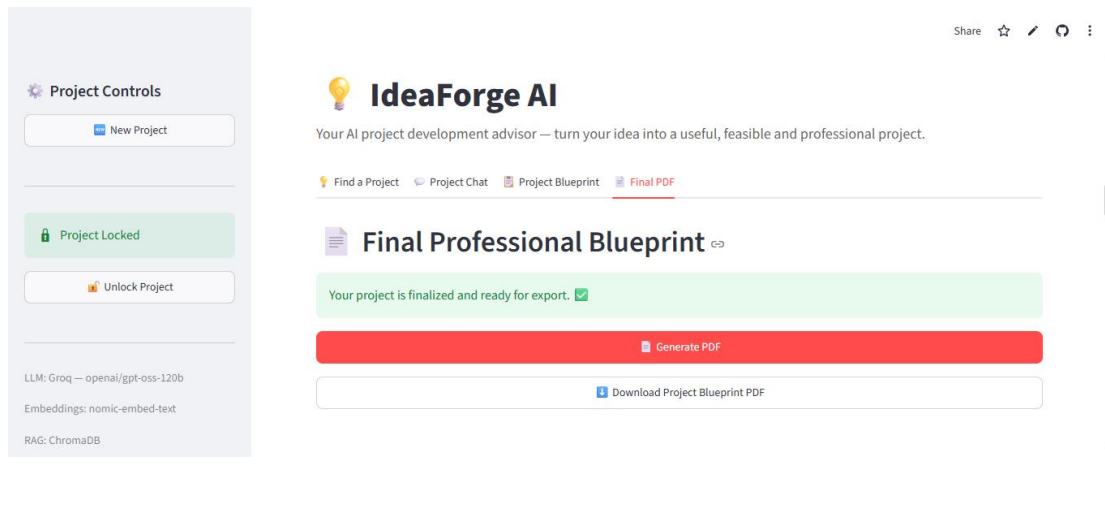
After a user requests a new feature, the feature and its description are added to the project.



Output 3 — Final Project Blueprint

Explanation:

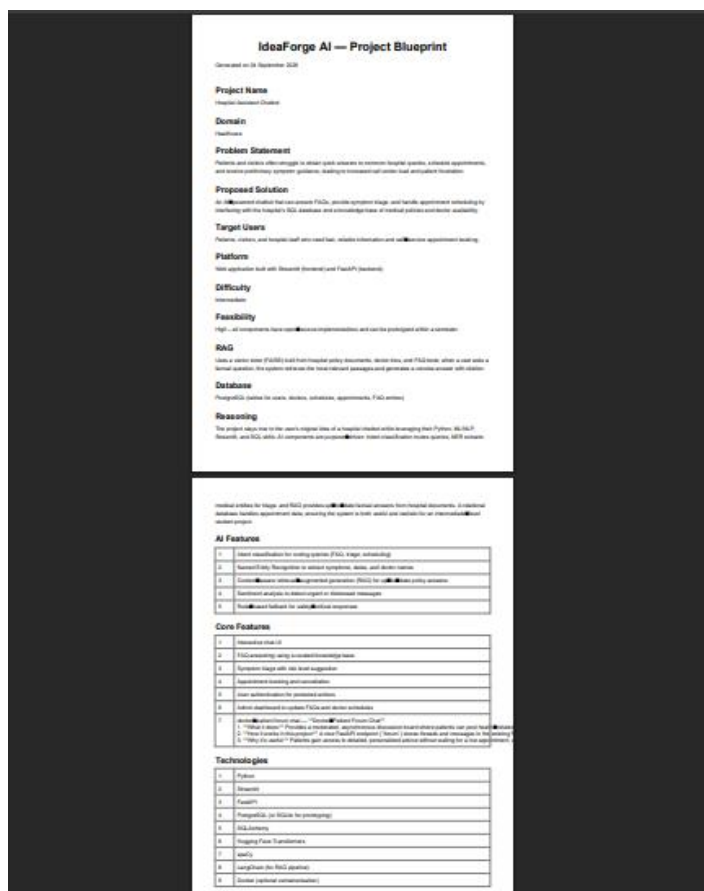
The finalized project is displayed as a structured blueprint containing the major project details.



Output 4 — Generated PDF

Explanation:

The final blueprint is converted into a downloadable PDF using ReportLab.



13. CHALLENGES & SOLUTIONS, LIMITATIONS

Challenges

1. LLM Integration

Integrating the language model with the application required handling API authentication and response generation.

Solution

The Groq API was integrated into the Python application using the Groq client.

2. Semantic Search

Keyword-based search alone is not sufficient for understanding similar project-related concepts.

Solution

Sentence Transformers were used to generate semantic embeddings and ChromaDB was used for vector retrieval.

3. Dynamic Feature Management

The project needed to respond to natural-language requests such as adding or modifying features.

Solution

Intent-based processing was implemented so that the application can identify the requested action and update the project state accordingly.

4. Local Development vs Deployment

The application worked locally with environment variables, but deployment required secure API-key management.

Solution

Streamlit Secrets was used to securely store the Groq API key during deployment.

5. Dependency Management

The application depends on multiple Python libraries.

Solution

A `requirements.txt` file was created to specify the dependencies required for deployment.

Limitations

The quality of generated recommendations depends partly on the language model.

RAG quality depends on the knowledge available in the vector database.

The application currently focuses primarily on project planning rather than automatically building the entire project.

The deployed version depends on access to the Groq API.

Advanced user authentication and persistent project history are not currently implemented.

14. FUTURE ENHANCEMENTS

Future versions of IdeaForge AI could include:

1. Project History

Allow users to save and revisit previous projects.

2. User Authentication

Provide individual user accounts.

3. Advanced Feasibility Analysis

Evaluate estimated cost, development time, technical complexity and resource requirements.

4. Automatic Code Generation

Generate project starter code based on the finalized blueprint.

5. GitHub Integration

Automatically create a GitHub repository and project structure.

6. Multiple LLM Providers

Allow users to select between different LLM providers and models.

7. Team Collaboration

Allow multiple users to work on the same project.

8. Cloud-Based Storage

Store project blueprints and user projects in a cloud database.

15. CONCLUSION

IdeaForge AI demonstrates how Generative AI, Large Language Models, Retrieval-Augmented Generation, embeddings and vector databases can be combined to solve a practical project-planning problem.

The system transforms a basic project idea into a structured project plan containing project analysis, target users, features, technology recommendations and other relevant information.

Its interactive chat functionality allows users to continuously refine the project without restarting the planning process. Features and project information can be modified through natural-language instructions.

The integration of Sentence Transformers and ChromaDB provides semantic knowledge retrieval, while Groq and the `openai/gpt-oss-120b` model provide natural-language understanding and generation.

Finally, the project blueprint can be converted into a downloadable PDF using ReportLab.

Overall, IdeaForge AI reduces the gap between **having a project idea and knowing how to approach building it**, making it particularly useful for students, beginners, developers and project mentors.

16. REFERENCES

Streamlit Documentation

<https://docs.streamlit.io/>

Groq Documentation

<https://console.groq.com/docs>

Sentence Transformers Documentation

<https://www.sbert.net/>

ChromaDB Documentation

<https://docs.trychroma.com/>

PROJECT SUMMARY

Project Name: IdeaForge AI

Domain: Generative AI / Large Language Models / RAG

Primary Language: Python

Interface: Streamlit

LLM: Groq — openai/gpt-oss-120b

Embeddings: Sentence Transformers — all-MiniLM-L6-v2

Vector Database: ChromaDB

PDF Generation: ReportLab

Deployment: Streamlit Community Cloud

Live Application:

<https://ideaforge-ai-a7aiugt57ktpn352ajhb2v.streamlit.app/>